# Logic

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# 1 Propositional Logic

Philosophers still debate about what a proposition really means. As a complete beginner, I mention some interpretations of it, but I by no means claim that this is the definitive definition.

**Definition 1.1 (Possible World)**

A **possible world** is a complete and consistent way the world is or could have been.

The **language** of propositional logic consists of just two things: propositions and connectives.

**Definition 1.2 (Proposition)**

A **proposition** does not have a formal definition, but we can describe it in the following ways.

1. They can be understood as an indicator function $f : W \to \{T, F\}^a$ that takes in a possible world and returns a truth value. We can also model it with the preimage of $f$ under $T$, i.e. the characteristic set of $f$.
2. They deal with **statements**, which are defined as declarative sentences having a truth value.

Propositions are either true or false.

[a] $T, F$ stands for True, False.

**Example 1.1 (The Sky Is Blue)**

The proposition that *the sky is blue* is represented as the function that returns $T$ for every possible world where the sky is blue.

These declarative sentences are contrasted with questions, such as *how are you doing?* and imperative statements such as *please run my models.* Such non-declarative sentences have no truth value.

A statement can contain one or more other statements as parts. For example, compound sentences form simpler sentences.

**Definition 1.3 (Connectives)**

Statements are combined with **logical connectives**.

| Connective | Symbols |
|---|---|
| AND | $A \wedge B$, $A \cdot B$, $AB$, $A\&B$, $A\&\&B$ |
| OR | $A \vee B$, $A + B$, $A \mid B$, $A \parallel B$ |
| NOT | $\neg A$, $-A$, $\overline{A}$, $\sim A$ |
| NAND | $\overline{A \wedge B}$, $A \mid B$, $\overline{A \cdot B}$ |
| NOR | $\overline{A \vee B}$, $A \downarrow B$, $\overline{A + B}$ |
| XOR | $A \veebar B$, $A \oplus B$ |
| XNOR | $A \odot B$ |
| IMPLIES | $A \Rightarrow B$, $A \supset B$, $A \to B$ |
| EQUIVALENT | $A \equiv B$, $A \Leftrightarrow B$, $A \leftrightarrow B$ |
| NONEQUIVALENT | $A \not\equiv B$, $A \not\Leftrightarrow B$, $A \not\leftrightarrow B$ |

Table 1: Logical Connectives and Their Symbols

**Definition 1.4 (Propositional Formula)**

Propositions, represented by letters and denoted **propositional variables**, along with these symbols for connectives, combine to make a **propositional formula**.

Propositional logic is not concerned with the structures of propositions beyond the point where they cannot be decomposed any more by logical connectives.

## 1.1 Arguments

At this point we may look at a set of propositions $P_1, \dots, P_n$ and try come to a logical conclusion $Q$. This is called an argument.

**Definition 1.5 (Argument)**

Let $P$ be a set of propositions, called the **premises**. Let $Q$ be a proposition, called the **conclusion**. Then an **argument** is an attempt to deduce $Q$ from $P$. It is written in the forms

1. If $P$, then $Q$.
2. $P \implies Q$

An argument is **valid** if and only if

1. It is necessary that if $P$ is true, $Q$ is true.
2. It is impossible for $P$ to be true, while $Q$ is false.

**Example 1.2 (A Valid Argument)**

The following is an argument.

1. If *it is raining*, then *it is cloudy*.
2. *It is raining*.
3. *Therefore it is cloudy*.

Logic in general aims to specify valid arguments. This is done by defining a valid argument as one in which its conclusion is a logical consequence of its premises. Determining whether a proposition is a a logical consequence of another proposition is the process of **deductive argument**, which has rules. These rules, called **rules of inference**, determines the "legal moves" from one or more premises to the conclusion. We give 2 familiar ones.

**Definition 1.6 (Modus Ponens)**

**Modus ponens** is a deductive argument form and rule of inference.[a] The argument states that given the premises

1. $P \implies Q$
2. $P$

Then our conclusion is $Q$.

[a] In some literature it is treated as an axiom, though most people think of it as a rule.

The next one is the familiar statement that a statement is equivalent to its contrapositive.

**Definition 1.7 (Modus Tollens)**

**Modus tollens** is a deductive argument form and a rule of inference. The argument states that given the premises

1. $P \implies Q$
2. $\neg Q$

Then our conclusion is not $P$.

# 2 First-Order Logic

In propositional logic, we deal with simple declarative propositions. **First-order logic** extends this by covering predicates and quantification. Let's motivate them.

We can think of predicates as properties. If we say *Socrates is a philosopher* and *Plato is a philosopher*, in propositional logic both these statements, represented as $P$ and $Q$, as utterances that are either true or false, and they are completely independent from one another. However, we may want to view them as an application of a predicate $*$ *is a philosopher* on the entities *Socrates* and *Plato*. This motives the formalism of the domain of discourse and the predicate.

**Definition 2.1 (Domain of Discourse)**

Given an individual $x$, its **domain of discourse** is the set over which certain variables of interest in some formal treatment may range.

**Definition 2.2 (Predicate)**

A **predicate** $P$ is a symbol that represents a property or a relation of a certain individual $x$ in a domain of discourse. Using predicates, $P(x)$ can be viewed as a proposition about the individual $x$.

Note that a predicate itself is not a proposition, since saying $*$ *is a philosopher* doesn't have any truth or false meaning to it, akin to a sentence fragment. But it is a placeholder $P(\cdot)$ upon which if an individual $x$ is put in, it makes sense to ask whether $P(x)$ is true.

**Definition 2.3 (Formula)**

A **formula** is a string of propositions, connectives, predicates, and variables $\phi$ that turns into a proposition once all free variables have been instantiated.

With predicates alone, all we have really done is add notational convenience. However, if we want to state a proposition not just about $x$, but its domain of discourse, then we can use quantifiers.

**Definition 2.4 (Quantifier)**

A **quantifier** is an operator that specifies how many individuals in the domain of source satisfy a proposition. The two most used quantifiers are

1. *Universal Quantification.* $\forall$, which means *for every.*
2. *Existential Quantification.* $\exists$, which means *there exists.*

These quantifiers are additional symbols in our language $\mathcal{L}$. If we add the equality symbol, we get first-order logic with equality.

**Axiom 2.0 (Equality)**

**Equality** is a primitive logical symbol which is always interpreted as the real equality relation between members of the domain of discourse. These equality axioms are:

1. *Reflexivity.* For each variable $x$, $x = x$.
2. *Substitution for Functions.* For all variables $x$ and $y$, and any function symbol $f$,

$$x = y \implies f(x) = f(y) \tag{1}$$

3. *Substitution for Formulas.* For any variables $x$ and $y$, and any formula $\phi(z)$ with free variable $z$, then
$$x = y \implies (\phi(x) \implies \phi(y)) \tag{2}$$

Symmetry and transitivity follow from the axioms above.

Ordinary first-order interpretations have a single domain of discourse over which all quantifiers range. **Many-sorted first-order logic**, or **typed first-order logic** allows variables to have different **sorts** or **types**, i.e. coming from different domains.

## 2.1 Exercises

**Exercise 2.1 (Shifrin Abstract Algebra Appendix 1.1)**

Negate the following sentences; in each case, indicate whether the original sentence or its negation is a true statement. Be sure to move the "not" through all the quantifiers.

1. For every integer $n \geq 2$, the number $2^n - 1$ is prime.
2. There exists a real number $M$ so that for all real numbers $t$, $|\sin t| \leq M$.
3. For every real number $x > 0$, there exists a real number $y > 0$ so that $xy > 1$.

*Solution.* Listed.

1. *Negation.* For at least one $n \geq 2$, the number $2^n - 1$ is composite (not prime). The negation is true. Consider $n = 4 \implies 2^4 - 1 = 15 = 3 \cdot 5$.
2. *Negation.* There exists no real number $M$ such that for all real numbers $t$, $|\sin t| \leq M$. The original is true. Pick $M = 1$, and by definition $|\sin t| \leq 1$.
3. *Negation.* For at least one real number $x > 0$, there exists no real number $y > 0$ so that $xy > 1$. The original is true. Given a real number $x > 0$, choose $y = \frac{1}{x} + 1$. Then,
$$xy = x\left(\frac{1}{x} + 1\right) = 1 + x > 1 \tag{3}$$
where the steps follow from the ordered field properties of $\mathbb{R}$.

**Exercise 2.2 (Shifrin Abstract Algebra Appendix 1.4)**

Suppose $n$ is an odd integer. Prove:

1. The equation $x^2 + x - n = 0$ has no solution $x \in \mathbb{Z}$.
2. Prove that for any $m \in \mathbb{Z}$, the equation $x^2 + 2mx + 2n = 0$ has no solution $x \in \mathbb{Z}$.

*Solution.* We prove by contradiction. Assume such a solution $x$ exists for odd $n$. We consider the two cases where $x$

1. is even.
$$\begin{aligned} x \text{ is even} &\implies x \equiv 0 \pmod 2 && (4)\\ &\implies x^2 + x \equiv 0 \pmod 2 && (5)\\ &\implies x^2 + x - n \equiv 1 \pmod 2 && (6)\end{aligned}$$
2. is odd.
$$\begin{aligned} x \text{ is odd} &\implies x \equiv 1 \pmod 2 && (7)\\ &\implies x^2 + x \equiv 1 + 1 \equiv 0 \pmod 2 && (8)\\ &\implies x^2 + x - n \equiv 1 \pmod 2 && (9)\end{aligned}$$

Both cases result in the quadratic expression lying in the equivalence class [1] and thus cannot be 0. This contradicts our assumption that it is a solution. We prove by contradiction. Assume a solution $x$ exists for odd $n$. Note that since $x^2 + 2mx + 2n \equiv x^2 \equiv 0 \pmod 2$, this implies that $x \equiv 0 \pmod 2$.[a] Therefore, we can write $x = 2x'$ for some $x' \in \mathbb{Z}$, our assumption is equivalent to the existence of $x'$. Substituting this gives

$$4x'^2 + 4mx' + 2n = 0 \iff 2x'^2 + 2mx' + n = 0 \tag{10}$$

Since $2x'^2 + 2mx'$ is even, $n$ must be even as well, which contradicts our assumption that $n$ is odd.

[a] This is true if we look at the contrapositive: $x \equiv 1 \implies x^2 \equiv 1$.

# 3 Second-Order Logic

First order logic can quantify over individuals, but not over properties. That is, while we can state something like

*There exists $x$ such that $x$ is a cube.*

we cannot quantify over a predicate. That is, the statement

*There exists a property $P$ such that a cube satisfies $P$.*

This statement does not make sense in first-order logic, but makes sense in second-order logic.